

Basic Concepts of Reliability

Introduction to Reliability Engineering. Failures and failure modes. Causes and mechanism of failures. Maintainability and availability.

Component Reliability and Hazard Models

Failure distributions. Constant and Time dependent hazard models. Exponential, Normal, Lognormal and Weibull failure models.

Modelling of System Reliability

Reliability block diagrams. Series and parallel configurations, Redundancies. Markov Chain.

Design for Reliability

Reliability allocation. Designing for higher reliability. Failure analysis: FMEA, FMECA, FTA. Reliability optimization.

Equipment Reliability Analysis

Type of failure data. Sources of failure data. Classification of failure data. Identification of Failure distributions. Goodness-of-fit Test. Analysis of equipment reliability.